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The Role of Network Centralization in Shaping Digital Sovereignty: An Analysis Under the DNS Lens

Andrei C. Azevedo | **Eder J. Scheid** | **Muriel F. Franco** | **Demétrio F. F. Boeira** | **Luciano Zembruzki** | **Lisandro Z. Granville**

Institute of Informatics (INF), Federal University of Rio Grande do Sul (UFRGS), Rio Grande do Sul, Brazil

Correspondence
Corresponding author Andrei C. Azevedo.
Email: acazevedo@inf.ufrgs.br

Abstract

Centralization of Internet-based services in few key players has been a topic of study in recent years. One of such services, the Domain Name System (DNS), is one of the pillars of the Internet, which allows users to access websites on the Internet through easy-to-remember domain names rather than complex numeric IP addresses. In this DNS context, the reliance on a small number of large DNS providers can lead to (a) risks of data breaches and disruption of service in the event of failures and (b) concerns about the digital sovereignty of countries regarding DNS hosting. As several essential services are provided through Electronic-Government (E-Gov), it is highly important to be able to measure the digital sovereignty of a nation and the impacts that the lack of such feature can bring to its citizens. This work approaches the issue of DNS concentration on the Internet by presenting a solution to measure DNS hosting centralization and digital sovereignty in different countries, such as Brazil, India, China, Russia, and South Africa. With the data obtained through these measurements, relevant questions are answered, such as which are the top-10 DNS providers, if there is DNS centralization, and how dependent countries are on such providers to manage domains using their country code Top-Level Domains (ccTLD). Future opportunities could investigate the impacts on sovereignty under the lens of other layers of the Open Systems Interconnection (OSI) Network Sovereignty representation model presented in this work.

KEYWORDS
Digital Sovereignty, Communication Protocols, DNS, Measurement, Internet Centralization

1 | INTRODUCTION

The centralization of the Internet and its underlying infrastructure is a digital sovereignty concern (Pohle & Thiel, 2020). Protocols and applications can render critical services and content unavailable in the event of failures caused by technical or political issues. One the most investigated protocols regarding centralization (Allman, 2018; G. C. M. Moura, Castro, Hardaker, Wullink, & Hesselman, 2020) is the Internet's Domain Name System (DNS). The DNS is a globally hierarchical naming mechanism that enables the association of networks, servers, and services to Internet Protocol (IP) addresses (Mockapetris & Dunlap, 1988). DNS enables, for example, accessing Websites through easy-to-remember domain names rather than IP addresses, meaning that *wikipedia.org* would be translated to `208.80.154.224`. The records that map domain names and IP addresses are maintained by authoritative DNS servers that provide authoritative and up-to-date records.

Because deploying a local DNS server requires technical expertise (Hao, Wang, Stavrou, & Smirni, 2015), companies not rarely have been delegating the task of maintaining their authoritative NameServers (NS) records to third-party DNS providers (*e.g.*, Cloudflare (Cloudflare, Inc., 2023) and Akamai (Schomp, Bhardwaj, Kurdoglu, Muhaimen, & Sitaraman, 2020)). Such a delegation, which has been increasing over the years (G. C. M. Moura et al., 2020), led to the current scenario where DNS resolution is concentrated on a small number of large providers. And, for the sake of the business model, each large DNS provider multiplexes its Information Technology (IT) or data center infrastructure among its client companies (Allman, 2018).

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As a result, DNS centralization inevitably leads to security and availability risks (Zembruzki, Jacobs, Landtreter, Granville, & Moura, 2020), such as user privacy and the inability to resolve domain names in case of an outage or service failure at one of the large providers. In addition, the overall digital dependency on a few IT service providers creates concerns regarding the *(a)* dependability and *(b)* digital sovereignty of countries (Roguski, 2019), especially considering compliance regulations, such as Europe's General Data Protection Regulation (GDPR) and Brazil's Data Protection Law (LGPD).

DNS centralization has been widely investigated in the literature. There exists a number of research efforts on assessing the degree of centralization in authoritative DNS servers (Zembruzki et al., 2020; Zembruzki, Jacobs, & Granville, 2022; G. C. M. Moura et al., 2020; Doan, Fries, & Bajpai, 2021), showing, for example, that popular domains share the same authoritative DNS servers. Thus, disruptions (*e.g.*, due to cyberattacks or sabotage) on DNS infrastructure providers could lead to collateral damages to multiple DNS domains.

Although this centralization aspect has been previously addressed, further research and actions on digital sovereignty implications is necessary considering such a DNS dependency (Kantas & Dekker, 2022). Analyzing digital sovereignty is crucial because it ensures a country's autonomy, control, and security over its digital infrastructure (Pohle & Thiel, 2020). Efforts to quantify the dependency of different countries on DNS providers are, thus, required to uncover possible sovereignty risks for the nations and their critical infrastructures (*e.g.*, healthcare, banking, and education sectors), too.

In this article, we investigate how country code Top-Level Domain (ccTLD) (*i.e.*, TLDs reserved for a country, sovereign state, or dependent territory) from two conglomerate of countries, *(i)* Brazil, Russia, India, China, and South Africa (BRICS) and *(ii)* the European Union (EU), are resolved and quantify their dependency on foreign DNS providers. For that, we define an approach to periodically collect measurements regarding `NS` records, `A` records, and `AAAA` records in order to find out and map the organizations responsible for managing such providers' infrastructure. These measurements use domains listed in the Tranco list (Le Pochat, Van Goethem, Tajalizadehkhoob, Korczyński, & Joosen, 2019). Thus, we also analyze how domains are managed and discuss the implications on regulations, compliance, and digital sovereignty under the DNS scope. The results show that DNS centralization is a reality and a key concern for digital sovereignty, especially for countries that do not have relevant DNS providers and rely on infrastructure providers from countries or companies with different regulations and interests. Complementing such findings, we present challenges and opportunities on the research of digital sovereignty focusing on the different layers of computer networks.

The rest of this article is organized as follows. In Section 1, we review background knowledge and discuss related work on DNS centralization and digital sovereignty. In Section 3, we introduce our *DNS Measurement* approach and its components, including implementation details. In Section 4, we present the evaluation and results, followed by a discussion and a list of research opportunities in Section 5. Finally, in Section 6, we close this article by presenting conclusions and discussions on future work.

2 | LITERATURE REVIEW

Due to the damage DNS centralization may bring to the Internet infrastructure (G. Moura et al., 2016), academia has been addressing and discussing such a topic in recent years. Efforts and observations found an alarming concentration of DNS traffic, with more than 50% of the observed traffic being handled by only 10 Autonomous System (AS) operators (Foremski, Gasser, & Moura, 2019). Further, this leads to efforts toward emerging topics to build a responsible Internet (Hesselman et al., 2020), which proposes more transparency and trust within networks independent of vendors and countries that run the underlying infrastructure. Thus, it is clear that companies from the technology and telecommunication sectors have a place to ensure secure communication and a key role in digital sovereignty.

2.1 | DNS Centralization

Despite the DNS being designed to be robust, with features such as delegations of names to multiple authoritative servers, redundancy, load balancing, and caching, operational practice frequently does not implement mechanisms to use these features properly. In the context of centralization, there have been concerns regarding the DNS and the neglect of its robustness mechanisms (Ramasubramanian & Sirer, 2005; Deccio, Chen, Mohapatra, Sedayao, & Kant, 2009). This can become a problem when

DNS operations rely on a single provider. Studies have shown that while DNS is engineered for high robustness, these features are often underutilized in practice (Allman, 2018). It is also showed that deploying a DNS infrastructure in a third-party provider can increase robustness, but this can also create single points of failure.

The issue of single points of failure becomes critical as DNS centralization progresses. The concern primarily stems from events that lead to DNS provider outages, as reported by studies. (G. Moura et al., 2016; Tarahomi, Holz, & Sperotto, 2023). The analysis of (Zembruzki, Jacobs, & Granville, 2022) sheds light on the DNS centralization problem by measuring and analyzing DNS authoritative resolvers for 19 TLDs. The authors show that more than 20% of all domains are hosted for the Top 5 DNS providers, and even more starling, roughly 80% of the whole IPv4 domain namespace under examination is hosted by the Top 100 DNS providers. In another study, (Kashaf, Sekar, & Agarwal, 2020) analyzed the influence of third-party dependencies on three distinct services: DNS, Content Delivery Network (CDN), and certificate revocation checks by Certificate Authority (CA). The authors reveal that 89% of Alexa's Top 100̧ websites rely on third-party DNS, CDN, or CA providers. The study also demonstrates that the use of third-party services is concentrated, with the top three providers of CDN, DNS, or CA services, thus affecting between 50% to 70% of the top 100 000 websites.

In the context of DNS traffic centralization, studies indicate that in certain ccTLDs, more than 30% of queries may be directed to a select number of large cloud providers. This concentration could have implications for DNS stability in different regions of the world (G. C. M. Moura et al., 2020). The work of (Radu & Hausding, 2020) analyzed DNS traffic centralization by measuring mobile devices using a tool created by the Tor project to identify censorship and anomalies in networks. The analysis revealed a reliance on public DNS resolvers in the first half of 2019, with more than 50% of DNS requests. The results also showed that Google and Cloudflare wield significant influence by controlling half of the overall market.

A distinct crucial concern is related to performance and how a centralized environment may negatively affect the response time of DNS in some regions of the globe (Doan et al., 2021; Silva, Franco, Scheid, Zembruzki, & Granville, 2024). Moreover, privacy is a main concern, as Internet Service Providers (ISP) typically operate DNS resolvers for their customers, meaning that they have access to users' DNS queries and can potentially monitor or manipulate the data, which is definitely a fact to watch. This centralization of DNS resolution can raise privacy and political concerns, mainly if ISPs engage in activities such as DNS filtering, censorship, or surveillance (Li et al., 2023).

Furthermore, another concern that DNS centralization may bring is security since cyberattacks are evolving and becoming more sophisticated (M. Franco et al., 2021), including those that target or explore DNS (*e.g.*, tunneling, amplification, and flooding) to cause technical, economic, and societal impacts (M. F. Franco, Granville, & Stiller, 2023). This security concern includes attacks on DNS authoritative servers (Jin, Tomoishi, & Matsuura, 2019; Aishwarya, Sannidhan, & Rajendran, 2014) and, also, availability of services worldwide, since the phenomenon of centralization is, in addition to being logical, also physical and geographical (Lee, Tan, Sekiya, Narishige, & Date, 2010).

2.2 | Digital Sovereignty Landscape

A relevant concept that emerges from the discussions on DNS centralization is the digital sovereignty. Although the notion of sovereignty related to nations in international relations is well established, its implications and dependencies related to the digital sphere are still being discussed. Since 2014, the relationship between sovereignty and the digital sphere has been a topic of increasing interest (Couture & Toupin, 2019), specially after Edward Snowden's leakage of documents revealing a mass surveillance program from the United States (US), which led to governmental attempts to take authority and control over how data from its citizens are managed, such as Europe's GDPR and Brazil's LGPD. Before the world's digital transformation, sovereignty was related to technology as a means to support national sovereignty with the development of commercial technology and industrialization (Couture & Toupin, 2019). Even though this idea is present in the concept of digital sovereignty, the usage of such a "catch-all" term now comprises several different aspects of sovereignty associated with data, governance, infrastructure and security (Hummel, Braun, Tretter, & Dabrock, 2021).

Hence, the terminology of digital sovereignty is applied as an umbrella term in the current literature that covers a wide range of facets. Different efforts have analysed the employment of terms related to sovereignty in the digital sphere – *digital sovereignty, cyber sovereignty, Internet sovereignty, technological Sovereignty* and *data sovereignty* – and to what aspects they are most related to (Hummel et al., 2021). Data Sovereignty, one of the pillars of sovereignty within the digital realm, is one of the most common and well-defined terms, that encompass the control of data flows by a nation and its subjection to a nation's jurisdiction (Maurer, Skierka, Morgus, & Hohmann, 2015; Couture & Toupin, 2019). Other characteristics present in the digital ecosystem are not well represented by the current available terminology in the context of sovereignty. Cyber

Sovereignty, technological sovereignty and digital sovereignty are often used as synonyms in a number of works, but they are also linked to specific attributes, such as Network Sovereignty considers the dependability of manufacturers from network operators (Janardhanan, Samonaki, Einar Heegaard, Kellerer, & Mas-Machuca, 2024). Therefore, it is clear that the different terms used in the literature to approach sovereignty in the digital are not well-categorized. Table 1 represents an overview of the coverage of digital sovereignty related terms in current literature.

TABLE 1 Overview of the Coverage of Digital Sovereignty Terms in Literature

Work	Data Sov.	Cyber Sov.	Technological Sov.	Network Sov.	Digital Sov.
(Maurer et al., 2015)	✓	✓	✓	✗	✗
(Couture & Toupin, 2019)	✓	✓	✓	✓	✓
(Floridi, 2020)	✓	✗	✗	✗	✓
(Pohle & Thiel, 2020)	✓	✓	✓	✗	✓
(Bauer & Erixon, 2020)	✓	✗	✓	✗	✓
(Hummel et al., 2021)	✓	✓	✓	✗	✓
(Barrinha & Christou, 2022)	✗	✗	✓	✗	✓
(Broeders, Cristiano, & Kaminska, 2023)	✓	✓	✓	✗	✓
(Hellmeer & Von Scherenberg, 2023)	✓	✓	✓	✗	✓
(Falkner, Heidebrecht, Obendiek, & Seidl, 2024)	✓	✗	✓	✗	✓
(Janardhanan et al., 2024)	✓	✗	✓	✓	✓

✓ Covered ✗ Not Covered

As can be seen, most of the available literature focuses on the coverage of data sovereignty, technological sovereignty and digital sovereignty. Cyber Sovereignty is also used as a synonym of digital sovereignty (Pohle & Thiel, 2020). This has also been studied by (Hummel et al., 2021), which reports that both Cyber and digital sovereignty are often related to notions of control and power in the context of sovereignty in the digital realm. The main difference is that the term Cyber Sovereignty is often employed in literature in the context of international relations, while digital sovereignty is used more frequently in the context of IT. Although the term *network sovereignty* is not commonly employed in the available literature, the literature still covers aspects related to network protocols and infrastructure, such as end-to-end encryption of communication data and localized routing (Maurer et al., 2015); thus, indicating that there seems to lack a clear categorization under the digital sovereignty umbrella.

The term technological sovereignty, which has been used since before the world's digital transformation, does not relate exclusively to digital technologies, but instead is a broader concept that encompasses the approaches of a nation to shape its development and use of technologies in a manner that reduces external dependencies, impacting also its economical and political sovereignty (Hellmeer & Von Scherenberg, 2023). Therefore, digital and data Sovereignty are facets of Technological Sovereignty, with the first being an umbrella term that comprises several characteristics of regulations and policies on the digital realm, and the latter being a subset of the first, more tied to data control, integrity, privacy and availability (Couture & Toupin, 2019). Hence, in this work, we consider the definition covered by the term digital sovereignty, as this this is the broader term used in the current literature when referring to characteristics of the digital realm that can impact on the sovereignty of a nation, such as network infrastructure and its related technologies (Hellmeer & Von Scherenberg, 2023).

Digital sovereignty refers to a nation's ability to control its digital infrastructure and digital technologies within its territorial borders (Aydn & Bensghir, 2019). It encompasses the idea that countries should be able to shape their digital policies, regulations, and frameworks to protect their national interests, security, and values in the digital realm. Digital sovereignty relies on specific aspects, such as data sovereignty for data protection and privacy regulations, domestic digital infrastructure, digital trade and economic policies, and Internet governance (Roguski, 2019). Studies have investigated sovereignty from different perspectives, such as the usage of the decentralization provided by blockchain technology (Scheid et al., 2021) as a potential ally for digital sovereignty (Manski & Manski, 2018). However, it is unlikely that fundamental changes will become a reality in the short term since, besides enormous technological efforts and associated costs, it depends on convergence between technical and political spheres.

Digital sovereignty is intrinsically linked with Electronic-Government (E-Gov) as services provided to citizens should be independent of foreign countries and highly resilient as they process and maintain personal and sensitive data of citizens.

(Sommese, Jonker, van der Ham, & Moura, 2022) assesses the DNS resilience of E-Gov in four countries (*e.g.*, The Netherlands, Sweden, Switzerland, and the US). The work measures the DNS structure of the list of domains used by the government of such countries and provides recommendations on how to improve the robustness of the DNS infrastructure of E-Gov services.

Besides underlying communications infrastructure, many scopes and challenges must be explored and addressed towards digital sovereignty. Examples include *(i)* data protection, *(ii)* technological independence (*e.g.*, 5G and mobile communications), *(iii)* cybersecurity, and *(iv)* commercial software applications. Besides all technical and economic arguments possible (M. F. Franco et al., 2023), cybersecurity is a key concern also for digital sovereignty because there is a considerable amount of money from a few groups and countries funding strategic cybersecurity companies worldwide. For example, in 2021, the US and Israel combined accounted for nearly 90% of all venture funding for cybersecurity companies (Crunchbase, 2021). Moreover, solutions towards data sovereignty (Hummel et al., 2021) are also under discussion, which involves strong data regulations that ensure compliance with privacy standards and user control over personal data.

The analysis of (Mitchell & Samlidis, 2022) focuses on the scope of data ownership and control. It investigates the risks and vulnerabilities of using cloud services for hosting Australian government data (*e.g.*, financial services, intellectual property, and administrative decisions) in different locations and regulations. The authors concluded that the scenario is manageable for the Australian case but argue that it is important to continue fostering and developing its cloud technology while negotiating and designing trade rules and regulations for data flows.

The discussion on digital and data sovereignty is also on the agenda of European industry and academia. For example, (Braud, Fromentoux, Radier, & Le Grand, 2021) proposes Gaia-X, an ecosystem composed of a trusted infrastructure and data space that allow secure data exchanges through federated nodes and actors. Furthermore, a federated architecture is explored to ensure privacy and data protection among all data providers, thus, resulting in an ecosystem that favors the digital sovereignty of conferments and companies.

Further, there are efforts to understand and reduce the dependency on physical devices, especially considering novel technologies (*e.g.*, 5G and 6G). In (Gábri & Hamuák, 2021), the authors discuss how the Slovak Republic is addressing such a concern in terms of allowing or prohibiting certain device suppliers from providing the 5G infrastructure in the country depending on security dimensions and taking the diversification of suppliers to avoid centralization in a single company.

In conclusion, digital sovereignty is a complex and evolving topic, and there are debates around its implementation and potential trade-offs. However, to the best of our knowledge, this is the first effort to analyze and discuss digital sovereignty of several countries based on the infrastructure of DNS providers of their ccTLDs. Hence, this article contributes to discussions about digital sovereignty under the technical lens of DNS and highlights opportunities for the analysis of digital sovereignty under the computer network lens.

3 | MEASUREMENT APPROACH

The approach consists of the mapping of lists of popular Internet domains (*e.g.*, based on the publicly available rankings) to its authoritative NSes and organizations behind providing such a service. This allows us to identify *(i)* who provides the correct IPs, *(ii)* which organization operates the NS infrastructure, and *(iii)* to which country and regulations the operator is subjected. For that, the approach combines information from domains (*e.g.*, `A`, `AAAA`, and `NS` records) and AS records provided by Internet registries (*e.g.*, LACNIC, RIPE, and ARIN).

An AS is a network of interconnected computing devices that operate under the same policy. It is often managed by a single entity (*e.g.*, ISPs or technology organizations) and is identified by an AS Number (ASN). Each AS manages one or more unique IP ranges, for example, *Wikimedia Foundation Inc.* has an ASN 14907 and manages the IP range `208.80.152.0/22` in the US and `185.71.138.0/24` in the Netherlands. Thus, it is possible to associate the IP of any NS to an AS and, consequently, to its operator and region.

Therefore, the approach is able to determine the entire flow from the domain name to the organization handling the AS that manages the IP of the associated NS. This allows us to understand the different points where centralization and digital sovereignty risks might occur. For example, the owner of an NS can tamper with the DNS records, while the AS operator can outage the communication.

In both scenarios mentioned, a clear DNS-related dependence can be identified on a few players that maintain the underlying infrastructure (*e.g.*, those that operate ASes and NSes). This makes the need to analyze such players and centralization a key pillar for discussing digital sovereignty.

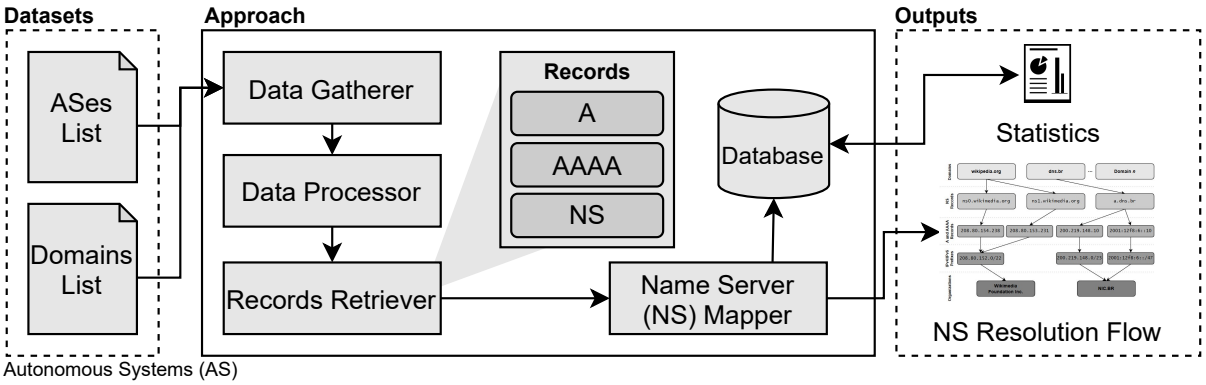


FIGURE 1 Overview of the Approach to Analysis Domains

Figure 1 depicts the components that are part of the approach and the flow of information between them. They are organized into three main groups, namely **Datasets**, **Approach**, and **Outputs**. Datasets containing information regarding *ASes* and a list of *Domains* are used as inputs.

The ASes responsible for each NS are defined using the list provided by the Center for Applied Internet Data Analysis (CAIDA). For that, it was used the network prefix mapping to AS (CAIDA, 2013) and the mapping of AS to organizations (CAIDA, 2014). This allows us to determine the AS, the organization managing the AS, and, thus, the country/region of DNS providers (based on the IP of the NS). For each measurement, an updated list of CAIDA is obtained by the *Data Gatherer* and processed so that the analysis rely on up-to-date information.

For the domains, the Tranco list (Le Pochat et al., 2019) is used as a dataset since it provides an updated source of the top 1 million Websites on the Internet based on popularity and access traffic. The list is updated considering different sources, *e.g.*, Alexa, SimilarWeb, and Moz; with the latest list used in the experiments (*cf.* Section 4) generated on June 16, 2023. This offers a reliable and transparent list that can be used to conduct research using popular domains. The *Data Gatherer* obtains the updated Tranco list for each measurement (using a diff approach to identify changes) and the *Data Processor* organizes the information of both ASes and domains to be used in further steps.

Next, the *Records Retrieves* analyzes each one of the 1 million domains and retrieves information regarding the `A`, `AAAA`, and `NS` records. For example, for the domain *wikipedia.org*, the `A` is `208.80.154.224`, the `AAAA` is `2620:0:861:ED1A::1` and the `NS` is `ns0.wikimedia.org`. This information is sent to the mapper to understand the entire path to resolve the DNS in order to build the *NS Resolution Flow* and to collects statistics (*e.g.*, organizations concentration, measurement errors, and identified IPs) for further analysis.

The *NS Mapper* receives the records regarding the domain and obtains the IP of the NS. This information is then used to map the IP to the correspondent AS managing it. For that, the `A` record can be used in case of an IPv4 prefix or the `AAAA` for IPv6. Finally, the organization name is obtained using a lookup to the CAIDA AS organization rank mapping dataset (CAIDA, 2014). Therefore, a complete analysis can be conducted to identify the nameserver region and relevant characteristics (*e.g.*, regulations and number of ASes being operated) so that insights from characteristics can be drawn.

The *NS Mapper* then stores information obtained in the *Database* and builds, as output, the *NS Resolution Flow*. This flow shows how the domain is resolved until discovering the organization or company that is managing the infrastructure, which is a point that may directly impact DNS resolutions in case of network disruption. Further, identifying NSes is crucial as they might tamper with DNS records, as they answer the requests in an authoritative manner.

Figure 2 illustrates a graph-like structure of the *NS Resolution Flow* for the domains *wikipedia.org* and *dns.br*. In the example, *wikipedia.org* has two `NS` records, *(i)* `ns0.wikimedia.org` and *(ii)* `ns1.wikimedia.org`, while *dns.br* has one, `a.dns.br`. This means these NSes are authoritative servers for these domains and are crucial to their operation.

This also applies to the organization that manages the IP addresses and advertises routing information of such servers (*i.e.*, their ASes). Such organizations, including the countries they are operating, are retrieved using the `A` and `AAAA` records of the NSes by identifying the resolved IPs using their prefixes and mapping them with the AS dataset list. Thus, in the example, *wikipedia.org* is managed by the Wikimedia Foundation Inc., located in the US, and *dns.br* is managed by NIC.BR, located in Brazil.

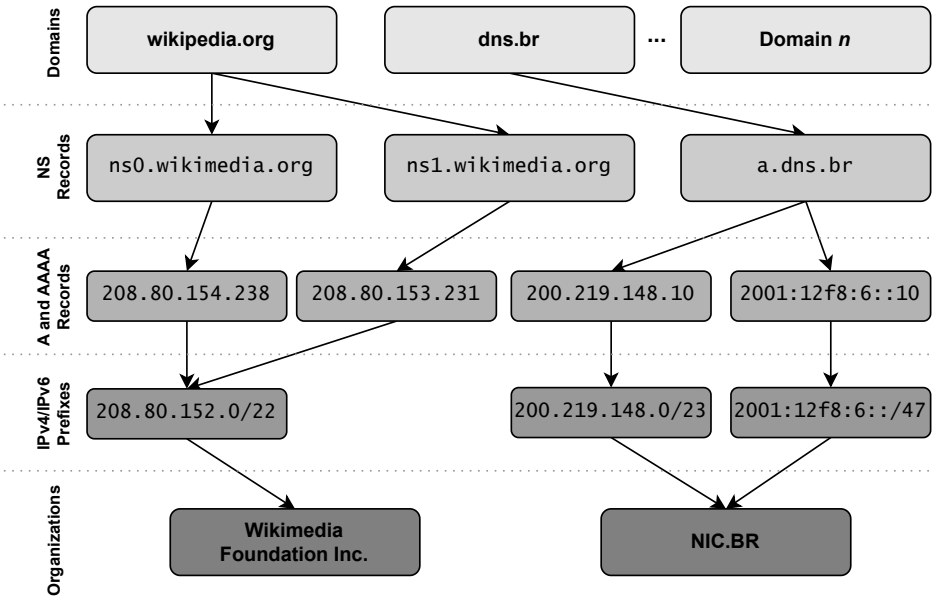


FIGURE 2 NS Resolution Flow Example

The implementation of the *DNS Measurement* approach and results of the evaluation of this article are available at (D. F. Boeira, Luciano Zembruzki, & Franco, 2023). Python was used to implement the approach's components, with the *dnspython* (Dnspython Contributors, 2020), a Python library to request and manipulate DNS records, being used to implement the *Records Retriever*. The *NS Mapper* connects with a *SQLite3* database to store the data required to build the *NS Resolution Flow*.

Further, statistics can be retrieved and processed from such a database. Table 2 provides examples of information collected using the approach. Thousands of entries (*i.e.*, domains analysis) were stored, following these metrics and organization, as CSV files for further analysis (*cf.* Section 4).

4 | EVALUATION AND ANALYSIS

The measurements considered all the 1 million domains from the Tranco list (Le Pochat et al., 2019), using only the pay-level domains filter, with the latest Tranco list used in the experiments generated on June 16, 2023. To infer the AS names and countries, the CAIDAs AS-to-organization dataset (CAIDA, 2013) was used. A six-core AMD Ryzen 5-5500U @ 2.1 GHz with 8 GB of RAM connected to the Internet using an Ethernet cable to maintain a stable network connection was used to conduct the measurements. Its operation system was a Debian 11 "*bullseye*" stable distribution.

It is essential to mention that during the experiments, not all domains from the Tranco list were resolved correctly (*e.g.*, DNS records not found or incorrectly configured), and their NS or ASN was not identified; thus, hindering the possibility of identifying the country where their DNS was managed. However, such limitation does not invalidate the results and contributions provided herein as for the digital sovereignty analysis the percentage of unresolved domains was < 5%.

4.1 | Identifying Top-10 DNS Providers

Table 3 lists the ranking, using 10 positions, of the DNS providers identified during the analysis of the centralization aspect of the DNS traffic. The position in the ranking is based on the number of domains that rely on such DNS providers during the indicated period. Three periods were defined, **Period 1** from 16/12/2022 to 23/01/2023, **Period 2** from 23/01/2023 to 13/02/2023, and **Period 3** from 13/02/2023 to 15/03/2023. As can be seen in the table, the ranking remained stable during these periods, and there was only one change, rows highlighted in gray in the table indicate a change in the ranking, where TIGGEE was the 6th during the first two periods but replaced MICROSOFT-CORP-MSN-AS-BLOCK as 7th in the third period.

TABLE 2 Example of Information Collected using the Measurement Approach

Information	Description	Example
Server Provider	Describe the nameserver allocated to resolve the domain	`ns3.google.com`
Organization Name	Identifies the organization owning the server	Google LLC
AS Number	Identifies the number of the AS managing the infrastructure of server provider	15 169
AS Country	Shows the country where the AS is operating	USA
AS Occurrences	Number of occurrences of the AS number controlling infrastructure of providers per country	30 812 (US) and 32 989 (RU) for BRICS
Centralization Percentage	The percentage that the AS occurrences represents in the data analyzed (*i.e.*, centralization)	35% (US) and 37% (RU) for BRICS

TABLE 3 Top-10 DNS Providers Identified

Position	Period 1	Period 2	Period 3
1st	CLOUDFLARENET	CLOUDFLARENET	CLOUDFLARENET
2nd	AMAZON-02	AMAZON-02	AMAZON-02
3rd	GODDADY-DNS	GODDADY-DNS	GODDADY-DNS
4th	ALIBABA-CN-NET	ALIBABA-CN-NET	ALIBABA-CN-NET
5th	GOOGLE	GOOGLE	GOOGLE
6th	TIGGEE	TIGGEE	MICROSOFT-CORP
7th	MICROSOFT-CORP	MICROSOFT-CORP	TIGGEE
8th	NSONE	NSONE	NSONE
9th	IONOS-AS	IONOS-AS	IONOS-AS
10th	OVH	OVH	OVH

Gray-highlighted rows indicate a change in the ranking.

Within this context, it was also investigated if the domains of such DNS providers (*e.g.*, *cloudflare.com*) were managed by them or if they relied on services from competitors. Table 4 presents the results of such investigation. The results indicate that not all DNS providers rely on their DNS services for their domains. For example, Amazon, the second largest DNS provider according to Table 3, uses Oracle's DNS services, and Godaddy, which employs its own DNS service but also relies on Akamai's DNS service. However, the majority of providers use their own DNS service.

TABLE 4 DNS Providers Domains and their Providers

Domain	DNS Provider	Country
cloudflare.com	CLOUDFLARENET	US
amazon.com	ORACLE-BMC-31898	US
godaddy.com	GODADDY-DNS, AKAMAI-ANS2	DE, NL
alibaba.com	ALIBABA-CN-NET	US
google.com	GOOGLE	US
tiggee.com	TIGGEE	US
microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US
ns1.com	NSONE	US
ionos.com	IONOS-AS	DE
ovh.com	OVH	FR

4.2 | Measuring DNS Centralization

Having identified the top-10 DNS providers that are responsible for hosting the highest amount of domains in the list, one question that arises is if there is an apparent centralization on those providers or if the DNS providing service is highly distributed to avoid Single Point of Failures (SPoF) or monopoly. To address this question, the concentration of domains resolved by the providers listed in Table 3 was measured from 16/12/2022 to 15/03/2023.

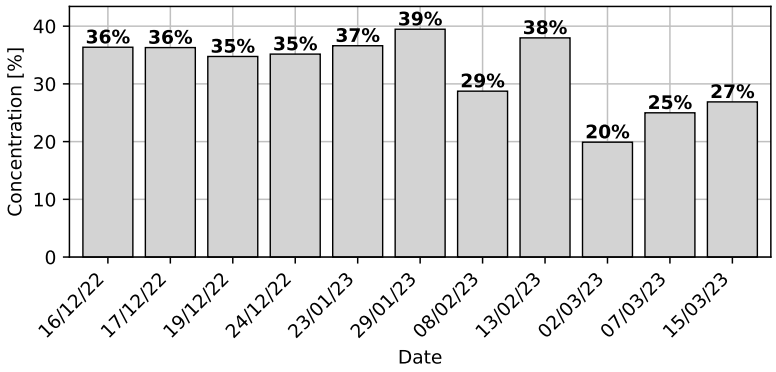


FIGURE 3 Concentration on Top-10 DNS Providers over 3 Months

Figure 3 depicts the results from the performed concentration measurements. In the figure, the *x*-axis represents the date on which the concentration percentage was calculated, and the *y*-axis represents the concentration in the top-10 providers. Considering the period, the average concentration was 30% of the measured domains. This means that, on average, 30% of the one million domains of the Tranco list (*i.e.*, 300 000 domains) had their DNS records hosted by the top 10 DNS providers (*cf.* Table 3). Further, considering that such a concentration peaked at 39% on 29/01/2023 and the fact that it was identified that around 3000 DNS providers were responsible for managing all of the one million domains, there is strong evidence that centralization in the DNS hosting industry is a reality.

4.3 | Analyzing Digital Sovereignty

Narrowing down the discussion on DNS centralization to a country-based analysis, it is possible to analyze countries' dependency on these providers and quantify how sovereign its Internet infrastructure is in terms of DNS hosting. For that, domains from the Tranco list were selected based on their ccTLDs (*e.g.*, `.br` and `.cn`) and grouped into their political conglomerates. In total, 91 286 domains from 95 792 domains using the BRICS and EU ccTLDs were resolved, and their DNS hosting organization identified. This represents 9.1% and 9.5% of the Tranco list, respectively. Russia's ccTLD (`.ru`) represented 59% of the resolved domains, approximately 54 168 domains. Results from such analysis categorized by these groups are presented in the following sections.

4.3.1 | BRICS Domains

BRICS represents a conglomerate of five major emerging economies, namely *(a)* Brazil, *(b)* China, *(c)* India, *(d)* Russia, and *(e)* South Africa, formed to promote inter-economic cooperation and inter-political discussions. As BRICS does not have an official ccTLD as Europe, the ccTLD for the BRICS are, respectively, *(a)* `.br`, *(b)* `.cn`, *(c)* `.in`, *(d)* `.ru`, and *(e)* `.za`.

Figure 4 depicts the results of the BRICS analysis. For each chart, the *x*-axis represents where the AS operates, and the *y*-axis represents the percentage of domains having their authoritative servers relying on such an AS. The countries are represented as Alpha-2 ISO country codes (International Organization for Standardization, 2023), and countries with less than 4% of domains were aggregated in the "Others" category. For example, in Brazil (*cf.* Figure 4a), there was a tie between `.br` domains that

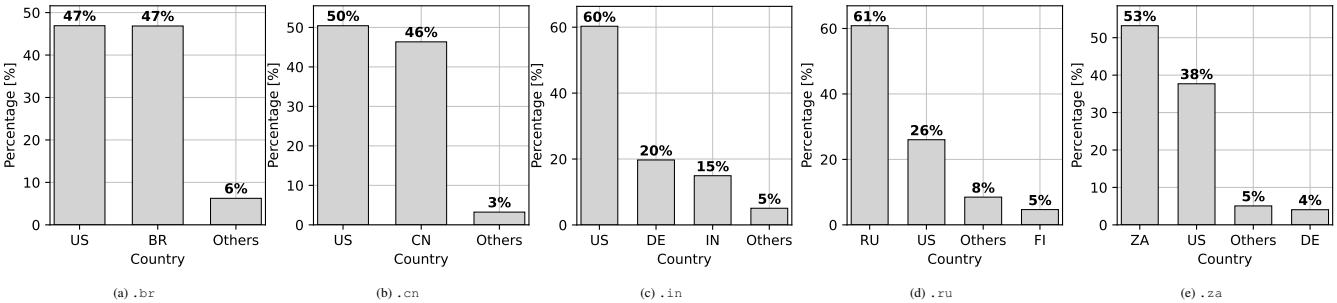


FIGURE 4 Results from the BRICS Domains Separated by ccTLD

relied on DNS providers from the US (*i.e.*, 47%) and domains that are provided by Brazilian-based companies (*i.e.*, 47%). The remaining share (*i.e.*, 6%) was located in other countries (*e.g.*, France and Germany).

It is possible to observe that US-based DNS providers, such as Cloudflare, Inc., Amazon.com, Inc., and Google LLC, represent a significant portion of the DNS hosting industry in the BRICS, with India presenting the highest dependence (*i.e.*, 60%) of the five nations. The exceptions are Russia (61%) and South Africa (53%), with most domains provided by national DNS companies (*e.g.*, Yandex.Cloud LLC for Russia and Xneelo (Pty) Ltd for South Africa). Thus, showing indications of concern regarding digital sovereignty.

Further, to have an overview of the digital sovereignty of the BRICS as a conglomerate, the five countries' results were aggregated and illustrated in Figure 5. Russia and the US appear to host the majority of the domains (*i.e.*, a total of 73%), followed by Brazil, China, and Germany. This behavior is logical considering the division of Figure 4. Therefore, showing a dystopian view of digital sovereignty, where the BRICS is subject to and dependent on the US regarding DNS regulations and infrastructure.

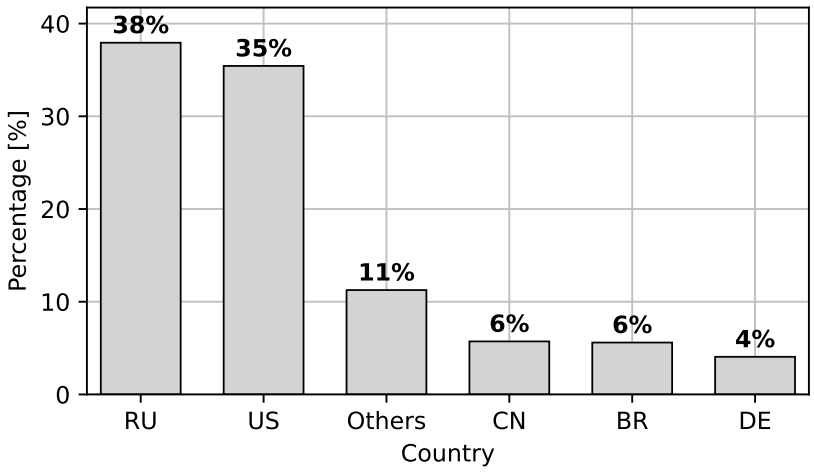


FIGURE 5 Results from the Aggregated BRICS Domains

4.3.2 | European Union

The EU is a political and economic union composed of 27 member states (*e.g.*, Portugal, Spain, France, Italy, Germany, and Belgium) located in Europe. For such countries, an initial analysis of the `.eu` ccTLD was performed because any person, company or organization within the EU may register domains with this ccTLD. Hence, it encompasses all European states, providing an overview of such an union. Figure 6 illustrates a different scenario than the one from the BRICS (*cf.* Figure 5),

where more countries share the DNS hosting infrastructure of the EU. Germany (*i.e.*, DE) represents a significant portion given its size and number of DNS hosting providers.

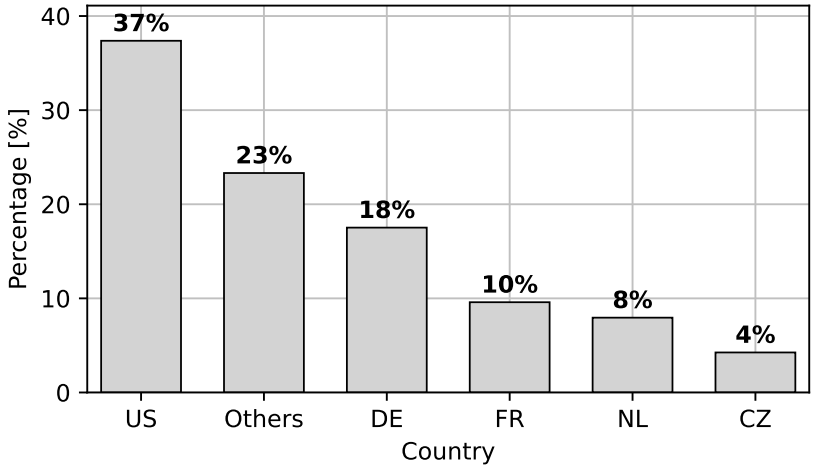


FIGURE 6 Results from the `.eu` Domain

However, the US also concentrates a significant portion of the DNS hosting industry for `.eu` domains. After Germany, France, and the Netherlands appear as major countries hosting DNS domains for Europe, this supports the data presented in Table 3, where OVH, a French cloud computing company, appears as the 10th DNS provider in the ranking. This concentration in a cloud provider might indicate that other services, besides DNS, are being hosted in France and the Netherlands, given the fact that such companies offer more services than DNS, such as virtual machines, Function-as-a-Service (FaaS), and web hosting that require a DNS provider.

In addition, extending the previous work performed by (D. Boeira, Scheid, Franco, Zembruzki, & Granville, 2024) and providing an in-depth perspective of Europe's digital sovereignty with respect to DNS hosting, 10 European ccTLDs were analyzed. The analysis followed the same methodology as the analysis of the BRICS domains performed in Section 4.3.1. Figure 7 depicts the results of the analysis performed using the following ccTLDs: Belgium (`.be`), Switzerland (`.ch`), Germany (`.de`), Spain (`.es`), France (`.fr`), Italy (`.it`), Netherlands (`.nl`), Poland (`.pl`), Portugal (`.pt`), and United Kingdom - UK (`.uk`). Although Switzerland (`.ch`) and the UK (`.uk`) are not part of the EU, Switzerland was included as it is part of the Schengen Area and the UK because it remains, even after its recent exit from the union, a key player in discussions regarding digital infrastructure and cybersecurity in the EU (Digibyte, 2023).

Based on the results of the in-depth analysis of selected European domains, a similar DNS hosting behavior can be identified for these domains compared to the `.eu` domain, where there was no country hosting the majority of domains; showing that there is no preference for foreign DNS providers. Furthermore, corroborating the data presented in Figure 6, more than 60% of German domains (*i.e.*, `.de` ccTLD) are hosted in-country. This indicates that companies and individuals prefer to rely on German DNS providers, highlighting the importance of local DNS hosting. Similarly, the same behavior is observed in France, Switzerland, Italy, the Netherlands, Portugal, and Poland. In contrast, the UK and Spain show that the majority of domains are hosted by US-based companies. However, closely followed by local or Europe-based companies, thus indicating a possible movement towards preferring local and European DNS hosting companies than foreign ones.

Lastly, comparing these results with those of the BRICS countries (*cf.* Figure 4), it is evident that European countries tend to prefer local DNS providers. This preference ensures that data is processed and stored within the country and that the DNS hosting service provider is bound by national laws (*e.g.*, the GDPR) and economic power (*i.e.*, countries with better infrastructure and investments tend to rely on internal infrastructure and providers). Additionally, local DNS providers can improve performance and reduce latency by being closer to users, which in turn improves user experience. This demonstrates that discussions on digital sovereignty (*cf.* Section 2.2) have a significant impact on the networking landscape of countries and the choice of service providers by companies.

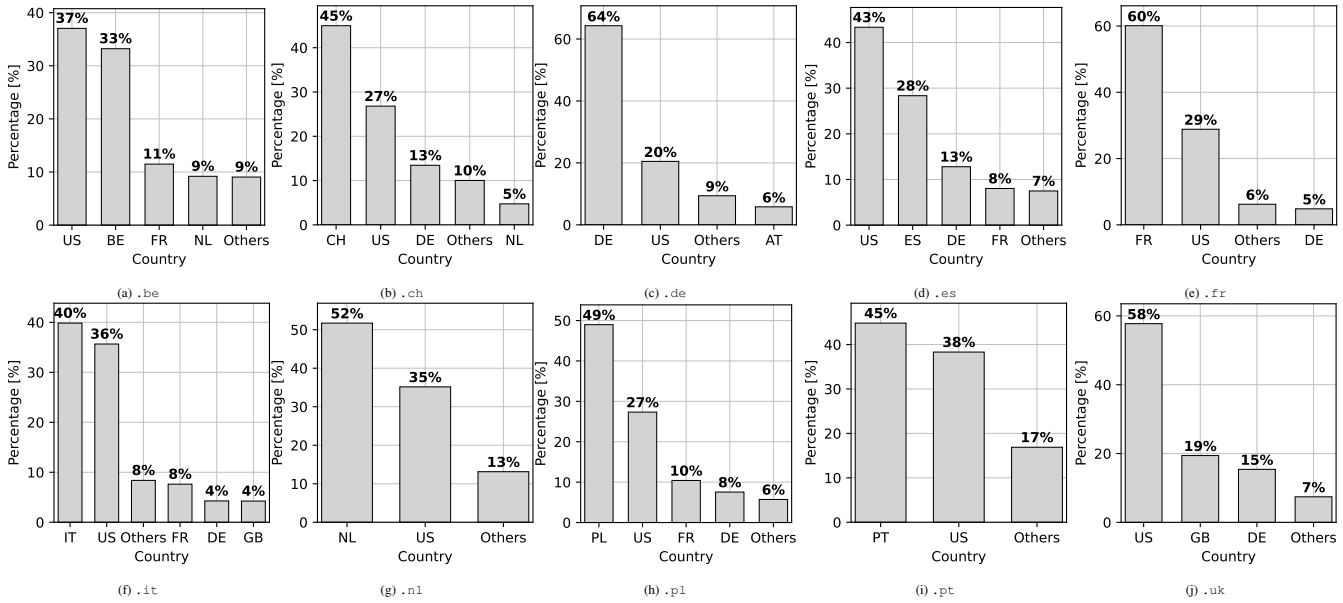


FIGURE 7 Results from Selected European Domains Separated by ccTLD

4.4 | Hosting Governmental Domains

One analysis dimension that is highly relevant concerning digital sovereignty and centralization is to investigate where restricted TLDs, such as `.gov`, are hosted. These domains are intended to be used only by federal government institutions (*e.g.*, security agencies and institutes). Thus, their DNS should be hosted within federal organizations to maintain critical services for citizens and control over the infrastructure during critical periods (*e.g.*, global conflicts, pandemics, or sanctions).

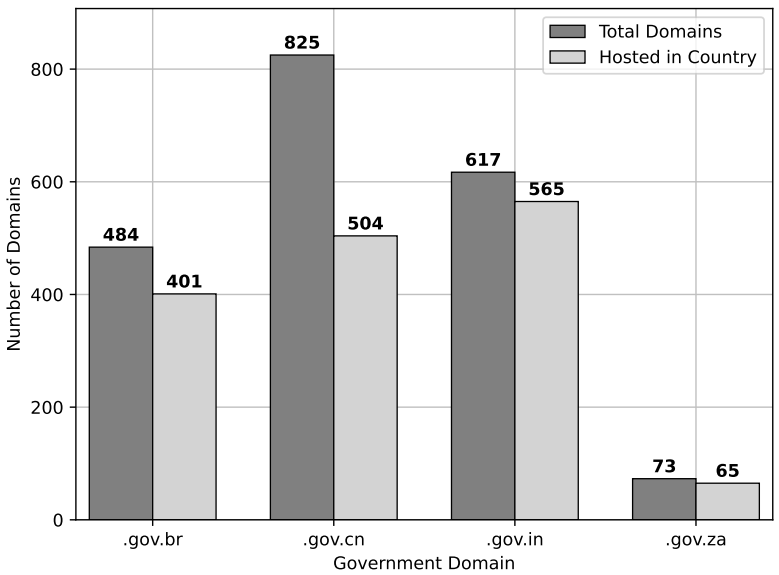


FIGURE 8 Results from the Analysis of the `.gov.` Domains

Figure 8 depicts the results from the analysis of the BRICS domains: `.gov.br`, `.gov.cn`, `.gov.in`, and `.gov.za`. Russia did not present `.gov` domains in the Tranco list; hence, it is not presented in the results. It can be seen that Brazil's

governmental domains are mostly resolved within Brazil, specifically in the Federal Data Processing Service (Serviço Federal de Processamento de Dados - SERPRO, in Portuguese), which is the biggest government-owned corporation of IT services in Brazil. Further, Indian and South African government domains are mostly hosted in their countries, with the National Informatics Centre (NIC) hosting most domains for India and the State Information Technology Agency (SITA) for South Africa. These results show a concern within BRICS about hosting governmental DNS domains for federal services within government organizations to avoid censorship, data leakage, and disruption of critical services.

5 | DISCUSSIONS

Different insights can be obtained from our experiments under different dimensions. From the technical dimension, we have shown that there is evidence of centralization on a few key players. Further, we showed that DNS centralization is economic in nature since big techs from developed countries lead the market. Moreover, several economic impacts (*e.g.*, business disruption and reputation harm) may happen in companies and governments in case of intentional or non-intentional disruption of the underlying DNS infrastructure. Our findings can also be explored from a legal dimension since digital sovereignty involves regulations and actions that can be done by policy-makers based on the technical analysis of the different protocols and dependence (*e.g.*, DNS and its centralization on a few companies and countries). The rest of this section provides a discussion on each dimension.

The key observations drawn from our experiments and analysis of digital sovereignty are presented in the rest of this section, followed by a discussion on the different challenges and opportunities to investigate digital sovereignty under the scope of computer networks and communication infrastructure.

5.1 | Key Observations

On the **technical** dimension, based on the results, it can be assumed that there is a clear indication of a DNS centralization, which can lead to a scenario where the Internet's infrastructure and management are directly dependent on a few key players (*e.g.*, governments and companies with different technical and political characteristics). This is not the best scenario since it can lead to the issues discussed in Section 1, such as security, assurance, and operational risks. Moreover, allowing such centralization in a given country, region, or company increases the risk of Internet censorship, as such a control can be achieved by injecting fake DNS replies to block access to certain content (Pearce et al., 2017). Thus, the DNS infrastructure and its distribution concentrated on a few authoritative servers may lead to Internet outages (due to misconfigurations) and Internet censorship, as the technical enablers for implementing this control are in place.

When discussing the economic dimension of DNS centralization, one point that relates is the possibility of DNS providers profiting from DNS lookup data. (Borgolte et al., 2019) advocates that DNS providers do not commercialize such information because of the potential consumer and regulatory backlash of such a monetization. However, suppose the DNS provider's centralization occurs in a country with not-so-well-defined regulations concerning commercializing user-sensitive data. In that case, further monopoly is risky as DNS lookup can be valuable for advertisement. Thus, monitoring and addressing DNS centralization and digital sovereignty is critical to tackling such an economic perspective. Further, most DNS providers (*e.g.*, Amazon, Google, and Microsoft) are also major cloud provider companies (Zembruzki, Sommese, et al., 2022), where their business is strongly tied to providing a reliable DNS infrastructure to access such cloud instances. However, such a combined service offering leads to a vendor lock-in issue (Opara-Martins, Sahandi, & Tian, 2016) and even further dependence on their infrastructure, in which clients are subject to such companies' pricing policies.

In addition to these possible economic impacts, DNS centralization has an economic motivation since big techs (often based in the US) offer DNS infrastructure, resolvers and associated services as part of their business core. In 2020, the DNS market was worth USD 372 million, and it is expected to be worth USD 862 million by 2025 (MarketsAndMarkets, 2023). This growth expectation is attributed to the increasing number of domain name registrations and Web traffic. Concerns about security, centralization, and digital sovereignty may be part of the marketing and product development strategies for DNS providers and big techs operating the underlying infrastructure.

Lastly, in the **legal and political** dimension, there are different efforts from the EU to strengthen its digital sovereignty, such as the GDPR for the idea of data sovereignty and the action plan for more digital sovereignty called by governments

of Germany, Estonia, Denmark, and Finland (Handelsblatt, 2021). Cybersecurity experts, entrepreneurs, and decision-makers also moved to the discussion to highlight the need to develop and promote digital infrastructures under European technological sovereignty (Rodosek, Broy, & Helmbrecht, 2021). Even though digital sovereignty is receiving much political attention around the world, discussions still need to evolve to find a common understanding to succeed in such dimensions.

In Brazil, the topic is being discussed among debates on different regulations that are required to increase national cybersecurity and digital sovereignty (Belli et al., 2023). Thus, as seen with these examples and discussions, digital sovereignty is a matter that many stakeholders (*e.g.*, governments, companies, and society) have to address from technical, economic, and legal perspectives. Otherwise, digital colonialism may become more prominent and dangerous in the following years, providing mechanisms to increase censorship and digital warfare.

Thus, as shown in this work and experiments, we advocate that the analysis and discussions on digital sovereignty under different lenses are needed. In parallel to the discussion on the centralization of protocols, such as DNS, different aspects, such as cybersecurity, regulations and investments for technology, and mobile communications and its vendors, must be investigated to lead the discussions of digital sovereignty.

As pointed out in (Nasir, 2023), data sovereignty is the enabler for organizations to achieve digital sovereignty fully. In this sense, companies and countries should ensure that DNS records and related data are processed and treated within country legislation and rules. Further, the technical aspect relates to striking a balance between companies and governments becoming dependent on in-house solutions that might become legacy systems and becoming dependent on single DNS provider services. The operational aspect relates to DNS services providing transparent information about DNS records and monitoring their infrastructure. Lastly, as discussed in Section 1, the DNS infrastructure is crucial for the availability of services and, in the case of DNS centralization, this infrastructure becomes a SPoF, which might affect not only a single service but its entire supply chain (*i.e.*, all services that rely on such DNS provider). Thus, critical services (*e.g.*, governmental and financial) must be resilient regarding DNS availability, allowing users and interested stakeholders to reach a specific service using human-readable names within the country's infrastructure.

5.2 | Challenges and Opportunities

Digital sovereignty relates to different layers (*cf.* Figure 9), which is adapted from (Nasir, Duncan, Helkenberg, & Claps, 2023); layers depicted in white color are not covered in the discussions of this article, while gray-highlighted layers were covered. Thus, the research and discussions presented in this article addressed the **data**, **technical**, **operational**, and **assurance** sovereignty layers within the *self-determination* aspect.

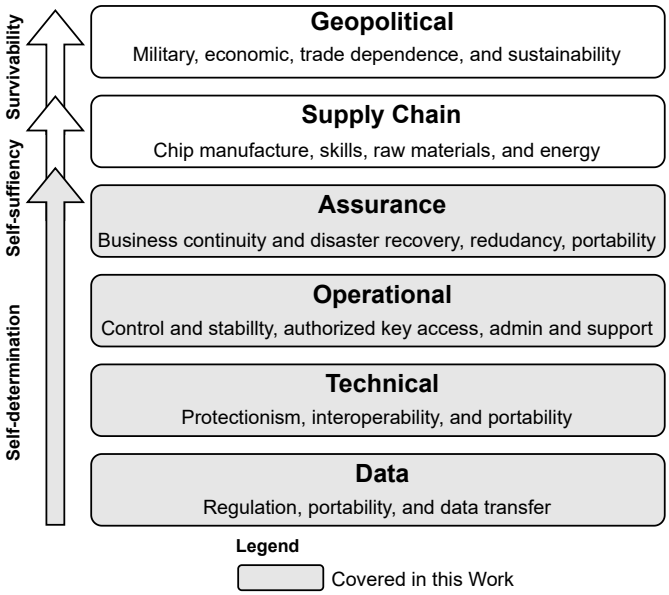


FIGURE 9 Digital Sovereignty Stack

However, each layer has its complexities and (lack of) standardization, making its analysis challenging since it may vary according to the observation and interpretation from multidisciplinary fields. For example, the analysis from a computer network perspective may see a traffic centralization, which can be described and perceived either positively or negatively based on its impact on digital sovereignty depending on where such a centralization occurs (*e.g.*, same country or a geopolitical partner). Therefore, there are opportunities for quantitative approaches that can provide measurable results to support the discussion on each layer of the digital sovereignty stack.

Furthermore, it is challenging to decouple digital sovereignty from policy-making (Belli & Jiang, 2024) since the digital goes in the direction of the sovereignty of nation-states. However, it is crucial to analyze digital sovereignty with different granularity towards measurable data and facts that can guide decision-making and discussions worldwide. One granular aspect is network sovereignty, where interoperability, centralization, and neutrality of communications must be investigated in an increasingly borderless cyberspace.

Network sovereignty covers, for example, traffic centralization, hosting of critical services and key technologies by specific countries, and dependence of manufacturers to operate communication infrastructure and services (Janardhanan et al., 2024). Figure 10 provides a representation of network sovereignty represented using the Open Systems Interconnection (OSI) model. Using OSI model as a conceptual framework, it is possible to highlight how network sovereignty can be addressed, considering the communication functions provided in each of the seven layers.

The bottom layers of OSI model (*i.e.*, Physical and Data Link) involve the acquisition and operation of equipment and building of the underlying communication infrastructure. This may impact the sovereignty in the context of regulations and vendor contracts. For example, in 2024, major telecommunications companies from Germany agreed to stop using critical components made by Chinese companies in their mobile communication infrastructure (Schuetze, 2024). In the case of replacement of 5G antennas only, this may already cost billions of Euros for Germany since more than 80 000 5G antennas made by Chinese companies are deployed in the European country (Morris, 2023). Therefore, additional analysis is needed to identify measurably the aspects of digital sovereignty, security gaps, and socioeconomic impacts that regulations and technological aspects may consider.

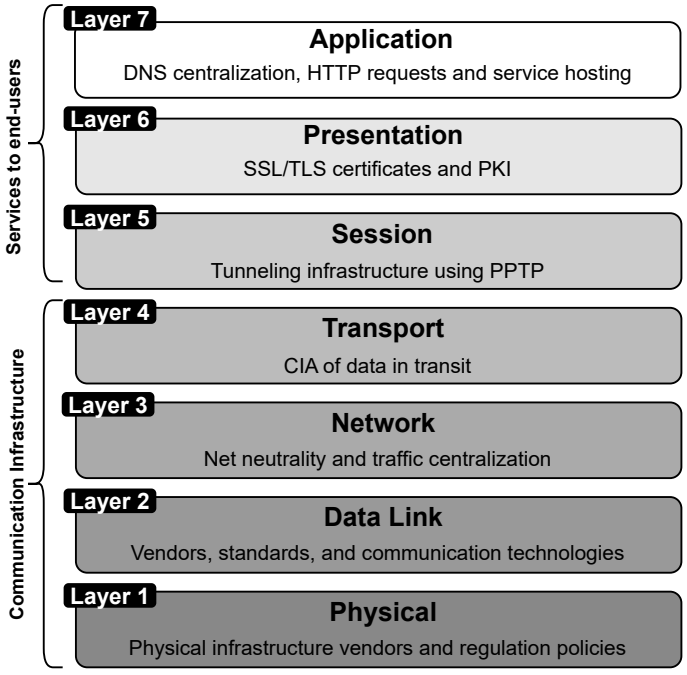


FIGURE 10 Examples of Network Sovereignty Represented under the OSI Model

Layers 3 and 4 (*i.e.*, Network and Transport) have been topics of investigation in the last decades due to the discussions towards net neutrality (Crowcroft, 2007) and traffic centralization. Recently, the centralization of blockchains has also been under discussion in terms of geographic distribution, routing centralization, and consensus power distribution (Sai, Buckley, Fitzgerald, & Gear, 2021). The centralization in the network layer is not seen necessarily as a negative aspect of digital sovereignty, especially because it may provide governance and enable the cooperation of larger and smaller players. However, there is a need to investigate how and when the centralization of traffic may become negative, such as in cases of having network traffic dropped or censored when depending on infrastructure and technology depending on external providers. Additionally, to collaborate with the discussion on more open and less vendor-dependent networking, there are also work advocating for novel approaches and concepts to propose and validate new models towards networks and communication systems with higher degrees of trust and sovereignty (Hesselman et al., 2020).

The upper layers of the OSI model (*i.e.*, Session, Presentation, and Application) are the ones where impacts on services to end-users are directly observed. For example, in the case of DNS centralization in external countries, it may cause disruption in critical services if the DNS resolution is not being provided by an intentional or non-intentional technical failure. The external dependence on Public Key Infrastructure (PKI) may also impact the confidentiality of communications since protocols such as Secure Sockets Layer (SSL) and Transport Layer Security (TLS) rely on the digital signatures generated by companies hosted in different countries. Therefore, the analysis of the centralization of the PKI and SSL/TLS protocols may reveal a lack of digital sovereignty for countries. Besides, there are also underlying technologies such as the Point-to-Point Tunneling Protocol (PPTP) that are commonly used to build Virtual Private Networks (VPN). The case of VPNs being hosted in infrastructure supported externally only may also harm the digital ecosystem of countries since companies and service providers frequently rely on VPNs to ensure connectivity and security functions.

This representation and discussion of digital sovereignty under the OSI model is an example of how a deeper investigation and understanding of different layers that ensure communication services can provide measurable and technical elements for the digital (and network) sovereignty debates. The analysis of critical services and infrastructure that society relies on to support their business and daily activities must be conducted; thus, making the political and regulation discussion supported by technical facts.

6 | CONCLUSION AND FUTURE WORK

The Domain Name System (DNS) infrastructure plays an essential role in the Internet access infrastructure by allowing content and services to be reached using easy-to-remember names (*i.e.*, domains). However, during its development, it was never imagined that such a system would become a market of global proportions. Thus, aspects such as its centralization and governmental regulations were disregarded. In this sense, given its central role in society and concerns regarding the level of control that DNS providers could enforce if the system becomes centralized, understanding and identifying DNS centralization is a key concern.

Thus, in this article, we presented an approach to measure DNS centralization and digital sovereignty based on DNS domain resolution. The approach relies on a list of 1 million popular domains (*i.e.*, the Tranco list) and, for each one, identifies the name server responsible for hosting the domain (*i.e.*, its authoritative server) and, based on its IP address, maps it to the Autonomous System (AS) managing the IP address.

Further, with the AS information, the approach identifies the country in which the AS is located to analyze which regulations the AS is subject to. Consequently, with that information, the approach infers the top-10 DNS providers, the percentage of centralization of the Tranco list in these providers, and the portion of domains that are managed within their country based on its country-code Top-Level Domain (ccTLD).

Results from the analysis show that most of the top-10 DNS providers identified in the Tranco list are in the US, with Cloudflare being the 1st DNS provider. Further, the analysis of how centralized the DNS hosting industry is revealed that the concentration of domains resolved in the identified top-10 providers peaked at almost 40%, which shows signals of centralization.

Lastly, the results of measuring digital sovereignty in Brazil, Russia, India, China, and South Africa (BRICS) and the European Union (EU) unveiled a scenario where a significant percentage of domains within these countries are not hosted by national companies but hosted on US-based organizations; exceptions being Russia and South Africa. Based on the results, it can be said that not only is DNS centralization occurring on the Internet as previous literature showed (*cf.* Section 1), but also

that countries are becoming less sovereign in terms of control over the national DNS infrastructure. We also highlight challenges and opportunities on how digital sovereignty can be analyzed from different perspectives, such as using the OSI model to show the dependency of communications and services to different protocols that may have centralized characteristics.

As future work, is it envisioned to *(a)* analyze DNS providers distribution with additional countries and address work limitations identified in Section 4, *(b)* investigate the centralization and potential impacts of additional protocols, such as the SSL/TLS certificates in digital sovereignty, *(c)* extend our measurement approach to support the analysis of the digital sovereignty under the OSI model, and *(d)* build a tool to analyze protocols centralization periodically. Furthermore, discussions on the dependency of technologies for communication and cybersecurity products must also be investigated since it may directly impact the applications and communication services running on the top of network infrastructure.

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